

Vikram Babar

Data Scientist

☎ 9322294771

✉ vikrambabar07@gmail.com

PROFILE SUMMARY

Versatile and Innovative Data Scientist with 3+ years of experience. Having ability to quickly adapt and apply my skills to new domains. I possess a strong track record of effectively communicating technical concepts to both technical and non-technical stakeholders, fostering collaboration and facilitating informed decision-making across departments.

KEY COMPETENCIES

- **Programming Languages:** Python, SQL(CRUD operations,3C).
 - **GenAi:** LLM, Langchain, Transformer models, Attention mechanism, Prompt design, Computer vision, Object detection.
 - **Machine learning:** Logistic Regression, Decision Tree, Random Forest, SVM, K-NN, Naive Bayes, K-Means, Ada-Boost, Ridge and Lasso Regression, Principal Component Analysis.
 - **Deep Learning:** Neural Networks, CNN, RNN, LSTM, Activation Functions, Back Propagation, Loss Functions, Optimizers, BERT.
 - **NLP:** TF-IDF, BOW, Word2Vec, Doc2Vec, Text understanding, representation & classification techniques.
 - **Cloud Platform:** AWS, Google Cloud Platform(GCP), FastAPI.
 - **Libraries:** Numpy, Pandas, Regex, TensorFlow, Matplotlib, Seaborn, SciPy, SciKit-Learn, Keras, BeautifulSoup, NLTK, SpaCy, GenSim, TextBlob.
 - **Maths and Stats:** Probability, statistics, linear algebra, Gradient Descent, P-Value, T-Test, Z-Test, ANNOVA test, Chi-Square Test, Info-Gain Test, Hypothesis Testing.
-

PROFESSIONAL EXPERIENCE

• Data Scientist | Veritas, Pune | 3.10 Years

Nov 2020 to Present Date

- **Project Title 1 :** RAG Application using Langchain and OpenAI(GPT-4).
 - **Domain :** GenAI and Natural Language Processing
 - **Description :** Development of Document Q&A system using Langchain and OpenAI, enhancing information retrieval from extensive documents. The project involved transforming text data into vector embeddings, storing them in a vector database, and retrieving relevant information to generate contextually accurate responses.
 - **Roles and Responsibilities :**
 - System Development: Designed and built the system, integrating LangChain with GPT-4.
 - Data Preparation: Converted textual data into vector embeddings for efficient retrieval.
 - Vector Database Management: Implemented and managed vector storage using FAISS.
 - Query Handling: Developed workflows to retrieve and integrate relevant context for query responses.
 - Optimized system performance to achieve 1.5-second response time and 93% accuracy.
- **Project Title 2 :** Real-time Object detection using YOLOv6 and OpenCV
 - **Domain :** Computer Vision and AI
 - **Description :** Development and deployment of a cutting-edge real-time object detection system using YOLOv6, a state-of-the-art deep learning model, within the domain of Computer Vision and Artificial Intelligence
 - **Roles and Responsibilities :**
 - Customized the YOLOv6 architecture, optimized hyperparameters, and fine-tuned the model for efficient inference on GPU hardware.
 - Achieved a mean Average Precision (mAP) of 0.75 on test data, demonstrating a 75% accuracy in identifying objects with at least 50% overlap.
 - Successfully integrated the model for live video inference, showcasing proficiency in deep learning, image processing, and model optimization techniques.
- **Project Title 3 :** Unveiling Business Insights: Sentiment Analysis of Electronics on a Ecommerce Platform for Consumer Sentiment Decoding

- **Domain :** E-commerce
 - **Roles and Responsibilities :**
 - Developed and Implemented machine learning models for sentiment classification.
 - Conducted data preprocessing, including text cleaning, tokenization, and feature engineering.
 - Evaluated and validated the model's performance for improved accuracy and generalization.
 - **Project Tittle 4 :** Advanced Phishing URL Detection Using Random Forest and Machine Learning
 - **Domain :** Cybersecurity
 - **Description :** Developed a phishing website detection system using machine learning algorithms, including Decision Tree, Random Forest, and Support Vector Machine. This project enhanced cybersecurity measures by effectively identifying phishing attacks with a low false positive rate.
 - **Roles and Responsibilities :**
 - Developed and optimized machine learning models to detect phishing websites, achieving a detection accuracy of 97.14% using the Random Forest algorithm, outperforming Decision Tree and SVM classifiers.
 - Implemented feature extraction techniques to analyze over 36,000 URLs, effectively identifying key characteristics of phishing sites, which led to a reduction in false positive rates to 2.61%
 - Evaluated model performance using a 90:10 dataset split, enhancing detection accuracy by 1.74% through robust validation.
 - **Process Executive | Nvidia, Pune | 11 Months | Feb 2018 to Dec 2018**
-

EDUCATION

- **B.E | 2017 | Pune University | Higher Second Class**
 - **12th | Kendriya Vidyalaya | CBSE | 77.83%**
 - **10th | Kendriya Vidyalaya | CBSE | 9.6 CGPA**
-

EXTRACURRICULAR ACTIVITIES

- Volunteer at **Pune Ploggers**, a City Cleanliness Initiative.
- Volunteer at **Robin Hood Army**, Multi-domain Community Service.